

Problem

Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: $Ax + By = D$ where D is the Greatest Common Divisor (GCD) of A and B. Given two positive integers A and B, find: $x \ y \ D = \text{gcd}(A, B)$ such that:

$Ax + By = D$

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$x \leq y$

Input Format

A B

Constraints

$1 \leq A, B \leq 10^9$

Output Format

$x \ y \ D$ D is $\text{gcd}(A, B)$ $Ax + By = D$ (x, y) satisfies the minimum-selection rules.

Submissions

Leaderboard

Discussions

Sample Input 0

4 6

Sample Output 0

-1 1 2

```
1 def gcd(a, b):
2     if b == 0:
3         return a, 1, 0
4     d, x1, y1 = gcd(b, a % b)
5     x = y1
6     y = x1 - (a // b) * y1
7     return d, x, y
8 a, b = map(int, input().split())
9 d, x, y = gcd(a, b)
10 p = b // d
11 q = a // d
12 k = round(-x / p)
13 best = None
14 for i in range(k - 2, k + 3):
15     xx = x + i * p
16     yy = y - i * q
17     cur = (abs(xx) + abs(yy), xx > yy, xx, y)
18     if best is None or cur < best[0]:
19         best = (cur, xx, yy)
20 print(best[1], best[2], d)
21
```

Line: 21 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

Download

1 4 6

Your Output (stdout)

1 -1 1 2

Expected Output

Download

1 -1 1 2